LIRAGLUTIDE

Daily SC injection

Pancreatitis

CHEMICAL NAME Arg34Lys26-(N-ε-(γ-Glu(N-α-hexadecanoyl)))-GLP-1[7-37] (acylated human GLP-1 analogue)

CLASS Glucagon-like peptide-1 (GLP-1) receptor agonist

TRADE NAMES Victoza (diabetes) · Saxenda (weight management) · Xultophy (with insulin degludec)

THERAPEUTIC USES / INDICATIONS

- Type 2 diabetes (with CV benefit — LEADER trial).
- Weight management (Saxenda, BMI ≥30 or ≥27 with comorbidity; adolescents ≥12 yr).

ADMINISTRATION

- **SC** once daily at any time (same time each day), independent of meals — abdomen, thigh or upper arm.
- Pen requires new needle each time; titrate weekly.
- Store unused pens in fridge; in-use up to 1 month at room temperature.

DOSAGES

- **Diabetes:** 0.6 mg once daily for 1 week, then 1.2 mg; max 1.8 mg.
- **Weight (Saxenda):** 0.6 mg, increase weekly by 0.6 mg to 3 mg daily.

PHARMACOKINETICS

- **Peak:** 8–12 h. **Duration:** 24 h.
- Albumin binding protects from DPP-4; proteolytic degradation.
- **t½:** 13 h.

NURSING MANAGEMENT

- Teach pen technique.
- Monitor weight, HbA1c, GI symptoms, hydration.
- Report severe abdominal pain (pancreatitis).
- Hold if vomiting with dehydration.

PHARMACODYNAMICS

- **GLP-1 receptor** agonist → glucose-dependent ↑ insulin, ↓ glucagon, delayed gastric emptying, ↑ satiety (central).
- Weight loss; ↓ cardiovascular events.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Nausea, vomiting, diarrhoea, constipation (early, dose-related).
- Decreased appetite.
- Injection-site reactions, headache.

ADVERSE EFFECTS (SERIOUS)

- Acute pancreatitis.
- Gallstones/cholecystitis.
- Dehydration → AKI (vomiting).
- Thyroid C-cell tumours (rodents); hypoglycaemia with SU/insulin; increased heart rate.

CONTRAINDICATIONS & CAUTIONS

- Personal/family history of medullary thyroid carcinoma or MEN2.
- History of pancreatitis (caution).
- Type 1 diabetes, DKA; pregnancy, breastfeeding.
- Severe gastroparesis.

KEY INTERACTIONS

- **Sulfonylureas, insulin:** hypoglycaemia; reduce doses.
- Delayed gastric emptying may alter absorption of oral drugs (e.g. warfarin — check INR).

PATIENT EDUCATION

- Inject once daily at the same time; nausea usually improves after a few weeks.
- Eat smaller meals; stop eating when full.
- Seek urgent help for severe stomach pain radiating to the back.
- Drink fluids to avoid dehydration.

KEY POINTS

- Daily GLP-1 agonist.
- Diabetes (1.2–1.8 mg) and weight loss (3 mg).
- Nausea; pancreatitis; gallstones.
- Avoid with MTC/MEN2 history.

SOURCES CHECKED **[620]** Liraglutide Biocon 6 mg/ml solution for injection in pre-filled pen (UK SmPC) **[621]** Saxenda, SC (US label)

Full references p. 432–457